BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 6197

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics - Paper: I - Objective (iv)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6197

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting

Note: SECTION-A

Q1. If α,β are roots of $x^2 - 4x + 5 = 0$ then αβ is:

(A) 2 (B) -4

(C) 5 (D) 4

Q2. The value of polynomial $5x^2 - 6x + 1$ at $x = -1$ is:

(A) -12 (B) 12

(C) 1 (D) 0

Q3. Partial fraction of $1/(x^3 + 1)$ is of the form:

(A) $A/(x+1) + B/(x^2-x+1)$ (B) $(Ax+B)/((x+1)(x^2+x+3))$

(C) $A/(x+1) + (Bx+c)/(x^2-x+1)$ (D) $(Ax+B)/(x^2+1) + C/(x+1)$

Q4. The multiplicative identity in complex number is:

(A) (1,0) (B) (0,1)

(C) (0,0) (D) (1,1)

Q5. If p is a logical statement then $p \wedge \sim p$ is always:

(A) Absurdity (B) Contingency

(C) Tautology (D) Conditional

Q6. If order of matrix A is m x n and order of matrix B is n x p, then order of AB is:

(A) m x n
(B) m x p
(C) p x n
(D) n x p

Q7. The inverse of a square matrix A does not exist if A is:

(A) Singular
(B) Non-Singular
(C) Unit
(D) Diagonal

Q8. If S is sample space, E is event of S, then P(E) is equal to:

(A) n(E)/n(S)
(B) n(S)/n(E)
(C) n(E).n(S)
(D) n(E)-n(S)

Q9. Expansion of (1 - 2x)^(1/2) is valid if:

(A) |x|<1
(B) |x|<1/3
(C) |x|<2
(D) |x|<1/2

Q10. In the expansion of $(a + b)^7$, the second term is:

(A) a^7
(B) $7a^6b$
(C) $7ab^6$
(D) 7b

Q11. Which of the given is not quadrantal angle?

(A) π/2
(B) 4π/3
(C) 9π/2
(D) 13π

Q12. A.M between 1/a and 1/b is:

(A) (a+b)/ab
(B) (a+b)/2ab
(C) 2ab/(a+b)
(D) ab/(a+b)

Q13. G.M between 1 and -16 is:

(A) -4
(B) 4
(C) ±4i
(D) 1/4

Q14. The number of ways of arranging 5 keys in a circular ring is:

(A) 24
(B) 12
(C) 6
(D) 5

Q15. Radius of e-circle opposite to vertex C is:

(A) Δ/(s-a)
(B) Δ/(s-b)
(C) Δ/(s-c)
(D) Δ/s

Q16. $\sin^{-1} A - \sin^{-1} B =$

(A) $\sin^{-1}[A\sqrt{(1-B^2)}-B\sqrt{(1-A^2)}]$

(B) $\sin^{-1}[A\sqrt{(1-B^2)}+B\sqrt{(1-A^2)}]$

(C) $\sin^{-1}[AB-(1-A^2)(1-B^2)]$

(D) $\cos^{-1}[A\sqrt{(1-B^2)}+B\sqrt{(1-A^2)}]$

Q17. Solution of equation $\text{cosec}\theta = 2$ in interval $[0,2\pi]$ is:

(A) $\pi/6, 7\pi/6$

(B) $\pi/6, 5\pi/6$

(C) $\pi/3, 5\pi/6$

(D) $\pi/3, \pi/6$

Q18. $\sin3\theta - \sin5\theta =$

(A) $2\sin4\theta\cos2\theta$

(B) $2\cos4\theta\sin2\theta$

(C) $2\sin4\theta\cos\theta$

(D) $-2\cos4\theta\sin\theta$

Q19. Range of $y = \cos x$ is:

(A) $-1 \leq x \leq 1$

(B) $-\infty < x < \infty$

(C) $-1 \leq y \leq 1$

(D) $-\infty < y < \infty$

Q20. In any triangle ABC, $(c^2 + a^2 - b^2)/2ac =$

(A) $\cos\beta$

(B) $\cos\alpha$

(C) $\cos\gamma$

(D) $\cos(\beta/2)$

Mathematics - Paper: I - Subjective

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Note: - Section B is compulsory. Attempt any three questions from section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Simplify by justifying each step $(1/4 + 1/5)/(1/4 - 1/5)$

ii. Separate into real and imaginary parts $i/(1+i)$

iii. $\forall z_1, z_2 \in C$, prove that $\overline{z_1+z_2} = \overline{z_1} + \overline{z_2}$

iv. Show A - B and B - A by Venn Diagram if $A \subseteq B$

v. Show that $\sim q \wedge (p \rightarrow q) \rightarrow \sim p$ is a Tautology.

vi. Prove that identity element in a group is unique.

vii. Find the inverse of $[[2i, i],[i, -i]]$

viii. Without expansion verify that $|[[-a, 0, c],[0, a, -b],[b, -c, 0]]| = 0$

ix. For what value of λ, the given system has non-trivial solution. {x + y + z = 0; 2x + y - λz = 0; x + 2y - 2z = 0}

x. Find three cube roots of -8

xi. Sum of a positive number and its reciprocal is 26/5. Find the number.

xii. Show that the roots of $px^2 - (p-q)x - q = 0$ are rational.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Resolve $1/(x^2 - 1)$ into partial fraction.

ii. Write only partial fraction form of $x^2/((x-1)^3(x+1))$. Not find the values of constants.

iii. Find the number of terms in A.P if $a_1 = 3$, d = 7 and $a_n = 59$

iv. If $a_{n-2} = 3n - 11$, find nth term of the sequence.

v. Find the G.M between -2i and 8i

vi. Find the Sum of an infinite geometric series 2+1+0.5+...

vii. Find the 8th term of H.P 1/2, 1/8, 1/8, ...

viii. How many 3- digit numbers can be formed by using each one of the digits 2,3,5,7,9 only once?

ix. Find the number of diagonals of a 8-sided figure.

x. Prove that the formula $3 + 3.5 + 3.5^2 + ... + 3.5^n = 3(5^{n+1} - 1)/4$ is true for n = 0,1

xi. Calculate $(0.99)^7$ by means of binomial theorem.

xii. If x is so small that its square and higher powers can be neglected then show that $(1-x)/\sqrt{(1+x)} \approx 1 - 3x/2$

Q4.. (Marks 18) Write short answers to any Nine parts. (9 x 2 = 18)

i. The pendulum of a clock is 20 cm long and it swings through an angle of 20° each second. How far does the tip of the pendulum move in 1 second?

ii. Find the value sin(19π/3)

iii. Prove the identity: $\cot^4\theta + \cot^2\theta = \operatorname{cosec}^4\theta - \operatorname{cosec}^2\theta$, where θ is not an integral multiple of π/2.

iv. Find the value of tan105°

v. Prove the identity $\cot\alpha - \tan\alpha = 2\cot 2\alpha$.

vi. Prove that $\sin(\pi/4 - \theta)\sin(\pi/4 + \theta) = 1/2 \cos 2\theta$

vii. Define period of a trigonometric function.

viii. At the top of a cliff 80 m high, the angle of depression of a boat is 12°. How far is the boat from the cliff?

ix. The area of triangle is 2437. If a = 79 and c = 97, then find angle β.

x. Prove that $r\, r_1\, r_2\, r_3 = \Delta^2$

xi. Find the value of $\tan^{-1}(-\sqrt{3})$.

xii. Define trigonometric equation.

xiii. Find the solution of the equation $\operatorname{cosec}\theta = 2$; which lies in $[0,2\pi]$

> Note: SECTION-C (Each question carries 10 marks)

Q5.. (Marks 5) (a) Use Crammer's rule to solve. $\{3x_1 + x_2 - x_3 = -4;\ x_1 + x_2 - 2x_3 = -4;\ -x_1 + 2x_2 - x_3 = 1\}$ (b) Solve the equation $x^4 - 6x^2 + 10 - 6/x^2 + 1/x^4 = 0$

Q6.. (Marks 5) (a) Resolve into partial fraction $x^2/((x+2)(x-1)^2)$ (b) Find n so that $(a^{n+1} + b^{n+1})/(a^n + b^n)$ may be A.M between a and b.

Q7.. (Marks 5) (a) Two coins are tossed twice each. Find the probability that the head appears on the first toss and the same face appear in the two tosses. (b) If x is very nearly equal 1, then prove that $px^p - qx\hat{}q \approx (p-q)x\hat{}(p+q)$

Q8.. (Marks 5) (a) If $\operatorname{cosec}\theta = (m^2+1)/2m$ and $0<\theta<\pi/2$ find the values of the remaining trigonometric ratios. (b) If $\sin\alpha = 4/5$ and $\cos\beta = 40/41$ where $0<\alpha<\pi/2$ and $0<\beta<\pi/2$ then show that $\sin(\alpha-\beta) = 133/205$

Q9.. (Marks 5) (a) Prove that $r_1r_2 + r_2r_3 + r_3r_1 = S^2$ (b) Show that $\tan(\sin^{-1}x) = x/\sqrt{(1-x^2)}$